

Experiment – 5

Definition:

Derive and reclassify fields for the telecommunications data to cleanse and enrich the dataset for model building.

Outcomes/Learning:

I learned how to use the Derive node to standardize inconsistent data and create new fields based on specific string functions.

Required Tool:

IBM SPSS Modeler 18.0

Working:

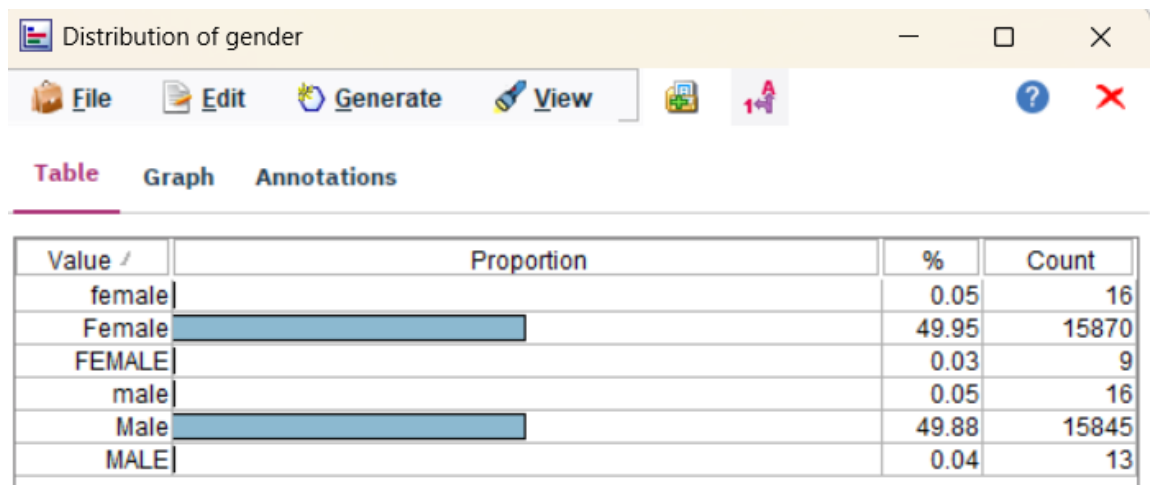
The process involves setting the working directory, opening a data stream, and using field operations to fix spelling inconsistencies in the 'gender' column.

Step 1: Open the start Stream.

From the File menu, click Open Stream, navigate to C:\Training\0A007108-Deriving_and_reclassifying_fields\Start, click unit 08 demonstration_1_start.str, and then click Open.

Step 2: Cleanse data for gender.

1. Run the Distribution node for gender, downstream from the Type node.



2. From the Field Ops palette, add a Derive node downstream from the Type node.

3. Edit the Derive node.

Settings Annotations

Mode: ☒ Single ☐ Multiple

Derive field:
Gender_OK

Derive as: Formula

Field type: <Default>

Formula:

L

4. Click the Launch expression builder L button to launch the Expression Builder.

5. click General Functions, and select the String group of functions from the list.

6. Under Function, select lowertoupper, and then click the Insert button to paste it into the expression area.

7. under Field, double-click gender, to add it to the expression area.

lowertoupper(gender)

General Functions

Function	Return
locchar_back(CHAR,N,STR...	Integer
lowertoupper(CHAR)	String
replace(SUBSTRING1,SUB...	String
replicate(N,STRING)	String
skipchar(CHAR,N,STRING)	Integer
skipchar_back(CHAR,N,ST...	Integer
soundex(STRING)	String

Fields

Type	Field	Storage
	customer_id	String
	gender	String
	age	Real
	postalcode	Real
	region	Real
	tariff	String
	connect_date	Date

8. add a distribution node, downstream from gender_ok.

9. Edit the distribution node, beside field select gender_ok list.

10. Click preview.

international_rate	voicemail	SMS	Gender_O
30	10	15	FEMALE
30	10	15	MALE
30	10	15	MALE
30	10	15	MALE
30	10	15	MALE
30	10	15	MALE
30	10	15	FEMALE
30	10	15	MALE
30	10	15	FEMALE
30	10	15	MALE
30	10	15	MALE
30	10	15	FEMALE

OK

11. Click Run.

Table	Graph	Annotations	
Value /	Proportion	%	Count
FEMALE		50.03	15895
MALE		49.97	15874

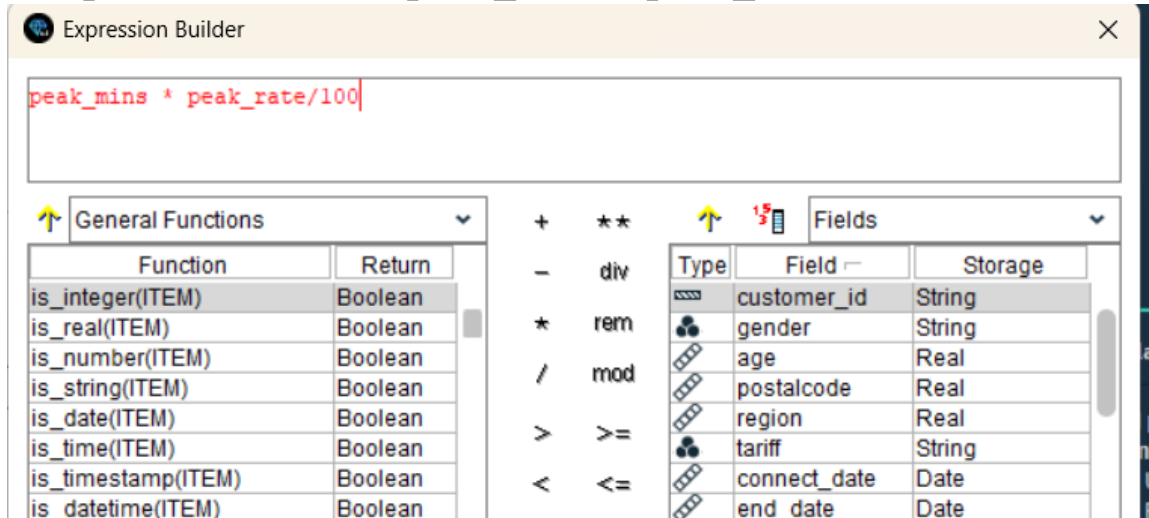
Step 3: Derive bill for peak, offpeak and total bill.

1 . From the Field Ops palette, add a Derive node downstream from the gender_OK Derive node.

2. Edit the Derive node

- Derive field: bill_peak.
- Derive as: Formula.

- **Expression Builder: $\text{peak_mins} * \text{peak_rate}/100$.**



3. Click Ok.

4. Click Preview.

voicemail	S...	Gender_OK	bill_peak
10	15	FEMALE	79.380
10	15	MALE	81.180
10	15	MALE FEMALE	61.290
10	15	MALE	77.940
10	15	MALE	48.510
10	15	FEMALE	74.700
10	15	MALE	75.150
10	15	FEMALE	82.890

5. Add a Derive node downstream from the `bill_peak` node.

6. edit it

- Derive field: bill offpeak.
- Derive as: Formula.
- Expression: $\text{offpeak_mins} * \text{offpeak_rate} / 100$.

7. Add a Derive node downstream from the `bill_offpeak` node.

8. edit it

- Derive field: bill total.
- Derive as: Formula.
- Expression $\text{bill_peak} + \text{bill offpeak}$.

8. Click Preview.

ler_OK	bill_peak	bill_total
LE	79.380	254.380
	81.180	92.180
	61.290	93.290
	77.940	107.940
	48.510	72.510
LE	74.700	135.700
	75.150	266.150
LE	82.890	91.890
	88.200	93.200
LE	67.230	122.230

OK

Step 4: Derive more fields for modelling.

1. Add a Derive node, downstream from the bill total node, and edit it Derive field:

`bill_total_greater_than_0`

- Derive as: Flag
- Field type: Flag (default value when the Derive type is Flag)
- True when: `bill total > 0`

Mode: ☒ Single ☐ Multiple

Derive field:

`bill_taotal_greater_than_0`

Derive as:

Field type:

True value: False value:

True when:

```
1 bill_total > 0
```

2. Click Ok.

3. Add a Derive node, downstream from the `bill_total_greater_than_0` node, and edit it as follows:

- Derive field: `segment`
- Derive as: Nominal
- Field type: Ordinal

4. Under Set field to. enter: 1, and under If this condition is true, enter bill_total <= 100.

bill total > 100 and bill total <= 200

bill total > 200.

5. Default value type undef.

Derive field:

segment

Derive as: Nominal

Field type: Ordinal

Default value: under

Set field to	If this condition is true
1	bill_total <=100
2	bill_total >100 and bill_total <=200
3	bill_total >200

OK Cancel Apply Reset

6. Click Ok.

7. Add a Derive node, downstream from segment node, and define it as follows:

- Derive field: discount.
- Derive as: Conditional.
- Field type: Continuous
- If: segment = 3.
- Then: bill_total *0.1

- Under Else, type 0.

Derive field:

discount

Derive as: Conditional

Field type: ⚡ <Default>

If:

1 segment = 3

Then:

1 bill_total * 0.10

Else:

1 0

OK Cancel Apply Reset

8. Click Ok.

9. Run the table node.

	K	bill_peak	bill_total	bill_total_greater_than_0	segment
463		78.210	227.210	T	3
464		60.030	116.030	T	2
465		77.400	185.400	T	2
466		58.590	145.590	T	2
467		52.920	95.920	T	1
468		70.200	119.200	T	2
469		65.340	191.340	T	2
470		75.690	80.690	T	1
471		78.930	92.930	T	1
472		60.480	86.480	T	1

Step 5: Cleanse data and reclassify fields for modelling.

- 1.add a Reclassify node, downstream from the discount node.
2. Edit the reclassify node.
3. Beside Reclassify into, select Existing field.
4. under reclassify field, select gender from the list.
5. click the get button.

Mode: ☒ Single ☐ Multiple

Reclassify into: ☐ New field ☒ Existing field

Reclassify field:

gender

New field name:

Reclassify1

Reclassify values:

 Get  Copy  Clear new  Auto...

Original value	New value
FEMALE	FEMALE
Female	FEMALE
MALE	MALE
Male	MALE
female	FEMALE

For unspecified values use: ☒ Original value ☐ Default value

OK Cancel Apply Reset

6. click Ok.

7. click preview:

Preview from Reclassify Node (60 fields, 10 records)

File Edit Generate

Table Annotations

	customer_id	gender	age	postalcode	region	tariff	connect_date	end_date
1	K338270	FEMALE	20.000	6599.000	3.000	CAT 100	2006-09-08	\$null\$
2	K342660	MALE	21.000	1635.000	1.000	CAT 100	2006-04-09	\$null\$
3	K342650	MALE	17.000	2149.000	1.000	CAT 100	2003-10-11	\$null\$
4	K342640	MALE	20.000	7788.000	4.000	CAT 100	2004-11-24	\$null\$
5	K342630	MALE	46.000	5223.000	3.000	CAT 100	2004-01-16	\$null\$
6	K342620	FEMALE	24.000	9143.000	4.000	CAT 100	2005-06-26	\$null\$
7	K342610	MALE	32.000	4223.000	2.000	CAT 100	2003-07-26	\$null\$
8	K342600	FEMALE	51.000	6165.000	3.000	CAT 100	2005-09-26	2007-06-
9	K342590	MALE	23.000	7210.000	3.000	CAT 100	2003-10-24	\$null\$

8. Click Ok.

9. add a reclassifying node, downstream form the Reclassify node.

Edit it:

-Reclassifying field: handset

-new field name: brand

10. click the get button.

Reclassify field:

handset

New field name:

brand

Reclassify values:

Get Copy Clear new Auto...

Original value	New value
ASAD170	ASAD
ASAD90	ASAD
BS110	BS
BS210	BS
CAS01	CAS

For unspecified values use: ☐ Original value ☒ Default value undef

OK Cancel Apply Reset

11. Click Ok.

12. add a Matrix node, downstream from the brand
Reclassifying node.

-Rows: handset

-column: brand

13. Click run.

Matrix Appearance Annotations

brand

handset	ASAD	BS	CAS	S	SOP	WC
ASAD170	2680	0	0	0	0	0
ASAD90	4355	0	0	0	0	0
BS110	0	5340	0	0	0	0
BS210	0	1391	0	0	0	0
CAS01	0	0	8	0	0	0
CAS30	0	0	2843	0	0	0
CAS60	0	0	506	0	0	0
S50	0	0	0	7076	0	0
S80	0	0	0	3442	0	0
SOP10	0	0	0	0	632	0

Cells contain: cross-tabulation of fields (including missing values)

Chi-square = 158,845, df = 55, probability = 0